

Dense CNN and IndRNN for the Sussex-Huawei Locomotion-Transportation Recognition Challenge

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ABSTRACT

The Sussex-Huawei Locomotion Challenge (SHL) 2021 organized at the HASCA Workshop of UbiComp 2021 is to recognize human activity by using GPS reception, GPS location, Wifi reception and GSM cell tower scans data. Compared with the previous challenge, this challenge is more difficult. In this paper, our team (NUC) summarize our submission to the competition. We propose a framework of deep learning including one-dimensional (1D) Dense CNN and Dense IndRNN networks to explore short and long-term spatial temporal information. The location and Wifi features are extracted and normalized to feed into deep learning networks. Finally, the decision fusion is utilized to improve performance.

CCS CONCEPTS

• Computing methodologies → Artificial intelligence.

KEYWORDS

SHL, activity recognition, Smartphone, deep learning

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1 INTRODUCTION

Human action recognition is to recognize human behavior by analyzing the movement information of human activities and has attracted increasing attention. It has a wide of potential applications

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such as medical diagnosis and monitoring, intelligent furniture and intelligent human-computer interaction. Depending on data, it can be divided into video based action recognition and inertial sensor based action recognition. In recent years, with the rapid development of computer science and sensor hardware, smart phones are equipped with more sensors, such as light, accelerometer, gyroscope, magnetometer and other sensors. The smart phones record human activities by collecting data through sensors. And smart phone-based human activity recognition has developed rapidly due to the advantages of small size, rich functions and convenient collection.

The SHL is to recognize the 8 activities (Still, Walking, Run, Bike, Car Bus, Train and Subway) using sensor data collected by smart phone [3, 14]. The SHL has been successfully held for three times (2018, 2019, 2020). The previous SHL uses motion sensors such as accelerometer, gyroscope to classify the modes of locomotion and transportation. Different from previous SHL, this SHL uses the GPS reception, GPS location, Wifi reception and GSM cell tower scans data and is more difficult. Since most satellites are asynchronous and the angle of the satellite is constantly changing. For Wifi data, it may have a fixed Wifi name in train and subway, but it has the different Wifi name for other activities. How to use these data and improve performance still remains an open problem.

With the promotion of SHL [12, 13, 15], many methods based on smart phone have been proposed in recent years. Depending on features, it can be divided into two categories: handcrafted features [5, 10] and deep learning features [1, 2, 6, 9, 16–19]. Handcrafted features based methods often use time domain and frequency domain features. These handcrafted features are fed into classical machine learning to classify activity. For example, the paper [10] use random forest as classifier to process handcrafted features such as linear velocity and average of gravity. Different from handcrafted features based methods, deep features based methods utilize deep learning model such as convolutional neural network (CNN) and recurrent neural network (RNN) to recognize human activity. In CNN-based methods, the data of different sensors are arranged into a 2- or 3- dimensional matrix and fed into CNN. For example, the temporal features of sensors can be directly fed into 1D CNN to explore the spatial-temporal information. However, this method only captures short-term temporal information by using length of kernel size. In order to capture long temporal information, RNN is employed such as IndRNN [8, 17], bidirectional LSTM [1].

This paper combines dense CNN and RNN to explore the short-term and long-term spatial-temporal information. Following with the DenseNet model, 1D DenseNet is constructed and dense IndRNN is used to extract more spatial information. The location and Wifi data are selected from four different kinds of sensor data. In order to improve recognition accuracy, decision fusion is utilized to fuse different scores from different models.

2 SHL DATASET

The data is divided into three parts: train, validate and test. The training data is collected by user1 in 59 days. The validation data is captured by 6 days with a mixture of user2 and user3. The testing data is also collected by user2 and user3 in 59 days. This SHL dataset includes GSM cell tower scans, GPS location, GPS reception and Wifi reception. The location data consists of accuracy of this location, latitude, longitude and altitude. The GPS reception contains number of satellite and each satellite includes four kinds of data (ID, SNR, Azimuth, Elevation). But these satellites are constantly changing for the earth, the angle of satellite is changing. Namely, an action is repeated at different time and we get different angles. The number of Wifi data is always variable and each Wifi contains BSSID, SSID, RSSI, frequency, and capabilities. For cell data, it consists of three types: LTE, WCDMA and GSM. Each type data has similar number of fields (isRegistered, ci, MCC, MNC, PCI, TAC, asuLevel, dBm, level).

3 PROPOSED METHOD

The framework of our proposed method is shown in Figure 1, and it includes three part: features extraction, training networks and decision fusion. First, the location and Wifi data are extracted and normalized from sensor data and then fed into dense CNN and IndRNN respectively to train model. Finally, the four scores generated from four networks are fused to recognize activity. In the following, the process to extract features is detailed

3.1 Features extraction

This SHL provides four types of data including GSM cell tower scans, GPS location, GPS reception and Wifi reception. In order to reduce impact of the useless data, the location and Wifi data are selected as our features. For location data, latitude, longitude and altitude features are used and motion features are extracted by subtracting the previous frame by the next one. For each kind of features in location it is normalized by using mean and variance. The mean and variance are obtained for each column of location data. For Wifi data, we get all SSIDs of wifi in test data. If one sample of train and val data does not containing the any SSIDs in test data, the sample will be deleted to optimize data. And these useful SSIDs are numbered as features and fed into deep networks.

3.2 Dense CNN and IndRNN

DenseNet [4] has been used in various computer vision tasks with promising performance. It has direct connections from any layer to all subsequent layers. This connection increases variation in the input of subsequent layers and improves efficiency. Motivated by DenseNet [4], we design a 1D DenseNet, by extending the DenseNet [4] from 2D spatial convolution to 1D spatial-temporal

convolution. The 1D DenseNet is used to extract features from sensor sequences. It consists of 1-D convolutional layers, pooling layers and Dense Block in [4]. The Dense Block receives the feature maps of all preceding layers. It can be defined as follows,

$$x_l = H_l([x_0, x_1, \dots, x_{l-1}]) \quad (1)$$

where $[x_0, x_1, \dots, x_{l-1}]$ refers to the concatenation of the feature maps produced in layers $0, 1, \dots, l-1$. H_l is a composite function of four consecutive operations: batch normalization (BN) followed by a rectified linear unit (ReLU); a 3×1 1D convolutional neural operation; batch normalization (BN) followed by a rectified linear unit (ReLU); a 3×1 1D convolutional neural operation. The hyper-parameters of the 1D DenseNet used in this paper are the same as those defined in [4] with growth rate $k = 32$ as shown in Table 1.

Table 1: 1D DenseNet architectures . Note that each “conv” layer shown in the table corresponds to BN-ReLU-Conv.

Layers	Output Size	DenseNet-121
Convolution	100	7×1 conv, stride 1
Dense Block (1)	100	$\begin{bmatrix} 3 \times 1 \text{ conv} \\ 3 \times 1 \text{ conv} \end{bmatrix} \times 6$
Transition Layer(1)	100	1×1 conv
	50	2×1 average pool, stride 2
Dense Block (2)	50	$\begin{bmatrix} 3 \times 1 \text{ conv} \\ 3 \times 1 \text{ conv} \end{bmatrix} \times 12$
Transition Layer(2)	50	1×1 conv
	25	2×1 average pool, stride 2
Dense Block (3)	25	$\begin{bmatrix} 3 \times 1 \text{ conv} \\ 3 \times 1 \text{ conv} \end{bmatrix} \times 24$
Transition Layer(3)	25	1×1 conv
	13	2×1 average pool, stride 2
Dense Block (4)	13	$\begin{bmatrix} 3 \times 1 \text{ conv} \\ 3 \times 1 \text{ conv} \end{bmatrix} \times 16$
Pooling	1×1	13 global average pool
Outputs		8 fully-connected, softmax

However, CNN can only capture short-term temporal information. In order to capture long-term temporal information, a deep dense IndRNN as in [7, 8] is utilized. The traditional RNNs where the recurrent inputs are processed with matrix product cannot effectively solve gradient exploding and vanishing problem, the recurrent inputs in IndRNN are processed with the Hadamard product as follows:

$$h_t = \sigma(Wx_t + u \odot h_{t-1} + b) \quad (2)$$

u is a vector and $\odot$ represents the Hadamard product. Compared with IndRNN, dense IndRNN following the structure in [4] explicitly processes all the features from all the previous layers and explores more effective information. The architecture of dense IndRNN is shown in Figure 2.

3.3 Decision fusion

In dense CNN and IndRNN, we don't use Softmax function. The average score fusion is used to improve performance. Different from directly adding four scores, the decision fusion includes two parts.

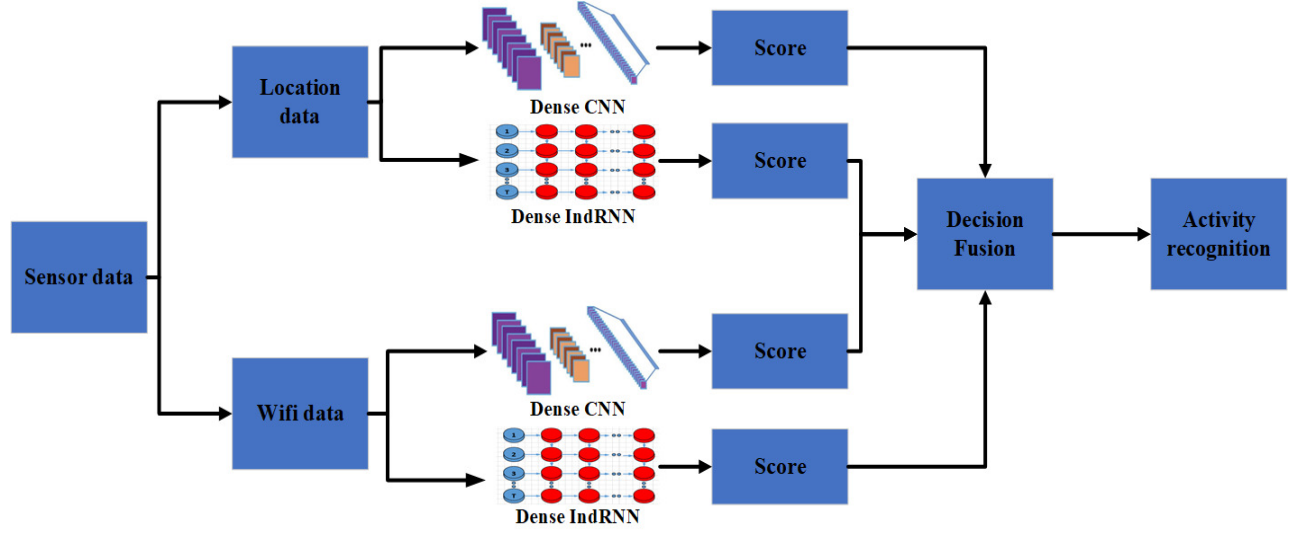


Figure 1: The framework of proposed method

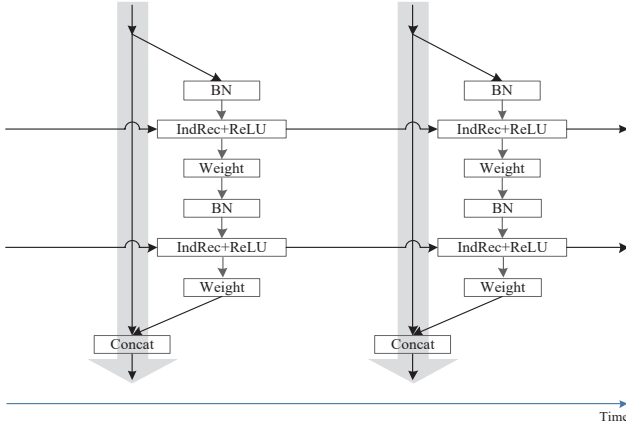


Figure 2: Dense IndRNN architecture [7]

First, the scores are fused using output of dense CNN and IndRNN for each data. After first fusion, part of Wifi scores is auxiliary to improve performance. According to confusion matrix of val data, Wifi data can effectively recognize train and subway activity. So scores of train and subway are fused with location scores. And other scores of Wifi are abandoned. In order to fusion decision, all test data from different sensors are synchronized according to temporal label of final test data.

4 EXPERIMENT

In experiment, length of location and Wifi data is 100 and 20 respectively. For dense CNN, the Adam optimization method with initial learning rate 0.002 is used. Weight decay is 0.0001 and dropout is 0.5 at last layer. For dense IndRNN, learning rate is 0.0002 and dropout is applied after the input (0.5), each dense layer (0.5), each bottleneck layer (0.1) and each transition layer expect (0.1 for the

still	0.463	0.207	0.014	0.064	0.110	0.090	0.005	0.046
walk	0.079	0.398	0.125	0.191	0.114	0.063	0.010	0.019
run	0.023	0.176	0.432	0.278	0.091	0.000	0.000	0.000
bike	0.027	0.074	0.128	0.411	0.196	0.054	0.111	0.000
car	0.034	0.086	0.007	0.111	0.644	0.071	0.039	0.008
bus	0.074	0.056	0.010	0.141	0.246	0.467	0.005	0.000
train	0.043	0.050	0.010	0.040	0.329	0.033	0.352	0.143
subway	0.015	0.008	0.000	0.000	0.000	0.000	0.053	0.925

Figure 3: The confusion matrix of the dense-CNN model using location data on the validation set

last and 0.3 for the others). For dense CNN and IndRNN, we adopt structure of 121 layers.

The results on the validation dataset using different networks are shown in Table 2. From the Table 2 it can be seen that location based results is better than Wifi data. And dense CNNs outperform dense IndRNN.

The confusion matrices with location data of dense CNN and IndRNN are represented in Figure 3 and Figure 4 respectively. We can see that the activity subway can be recognized effectively using location data. The confusion matrices with Wifi data of dense CNN and IndRNN are represented in Figure 5 and Figure 6 respectively. From the Figure 6, it can be seen that activity train and subway are easily distinguishable due to fixed Wifi name in train and subway. It is not available for other activities.

still	0.283	0.223	0.000	0.030	0.124	0.185	0.090	0.065
walk	0.124	0.439	0.082	0.174	0.075	0.004	0.068	0.034
run	0.017	0.167	0.683	0.122	0.011	0.000	0.000	0.000
bike	0.089	0.039	0.151	0.510	0.170	0.036	0.005	0.000
car	0.018	0.092	0.003	0.044	0.573	0.134	0.126	0.011
bus	0.072	0.054	0.000	0.178	0.245	0.402	0.049	0.000
train	0.035	0.063	0.005	0.023	0.535	0.025	0.134	0.179
subway	0.061	0.000	0.000	0.000	0.212	0.000	0.000	0.727

Figure 4: The confusion matrix of the dense-IndRNN model using location data on the validation set

still	0.514	0.124	0.035	0.047	0.019	0.110	0.084	0.068
walk	0.338	0.247	0.023	0.127	0.012	0.146	0.056	0.051
run	0.015	0.642	0.058	0.270	0.000	0.015	0.000	0.000
bike	0.012	0.127	0.000	0.000	0.006	0.694	0.145	0.017
car	0.141	0.340	0.000	0.153	0.036	0.075	0.248	0.007
bus	0.078	0.207	0.043	0.027	0.010	0.476	0.068	0.092
train	0.081	0.177	0.011	0.081	0.000	0.140	0.441	0.070
subway	0.036	0.009	0.002	0.173	0.002	0.000	0.009	0.769

Figure 5: The confusion matrix of the dense-CNN model using wifi data on the validation set

still	0.287	0.019	0.004	0.228	0.109	0.218	0.083	0.052
walk	0.222	0.258	0.001	0.129	0.067	0.209	0.089	0.026
run	0.000	0.007	0.000	0.268	0.056	0.613	0.056	0.000
bike	0.070	0.000	0.000	0.006	0.000	0.919	0.006	0.000
car	0.068	0.117	0.110	0.218	0.119	0.031	0.225	0.112
bus	0.192	0.065	0.027	0.031	0.035	0.550	0.067	0.033
train	0.000	0.000	0.000	0.137	0.068	0.058	0.737	0.000
subway	0.000	0.119	0.000	0.008	0.000	0.000	0.000	0.873

Figure 6: The confusion matrix of the dense-IndRNN model using wifi data on the validation set

5 COMPUTATIONAL RESOURCES

Software: Pytorch 1.8

Table 2: Accuracy of the proposed model on the validation set.

Data	networks	accuracy
location	Dense CNN	47.91%
location	Dense IndRNN	44.07%
Wifi	Dense CNN	35.79%
Wifi	Dense IndRNN	34.11%

GPU: GTXtitan

CPU: Intel Core E5-2640 V4

Model size: CNN:17MB; IndRNN:10MB

Training time: 9 hours for each model

Testing time: 5 minutes for each model

6 CONCLUSION

This paper presents a deep framework including dense CNN and IndRNN to recognize human activity. The framework effectively takes the advantages of dense CNN and IndRNN models in exploiting short-term spatial-temporal information and mining long term spatial-temporal information, respectively. The features of location and Wifi are generated to be fed into the deep framework. The effective average-score fusion is employed to improve recognition accuracy. The recognition result for the testing dataset will be presented in the summary paper of the challenge [11].

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